

Corentin Le Guen

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Senior Backend Developer & Data Engineer

Software developer with experience in the banking industry, specialized in creating robust and scalable solutions for complex business needs. Passionate about leveraging technology to solve challenging problems and create value. Expertise in back-end development, real-time data processing, cloud deployment, and data analytics. Modernized architectures, developed critical applications, and implemented cutting-edge tools to support high-frequency trading and financial analysis. Combines technical proficiency with a strategic mindset to drive performance and innovation. Thrives in collaborative environments, delivering projects and mentoring others.

Technologies

Back-end: Python, Java, Kotlin, Q/KDB+, Bash

Databases: Cassandra, SQLite3, MySQL, PostgreSQL, MongoDB, Redis

Testing: Gatling, Mockito, Junit, Cucumber, Sonar

Tools & Frameworks: Flask, FastAPI, Splunk, DataDog, Maven, Git, GitLab / GitHub / Bitbucket, Postman, PyCharm, IntelliJ, Docker, Jenkins

Professional Experience

Alithya

Juin 2025

Senior Developer & Tech lead

Client: Desjardins

- Designed and implemented an architecture following latest software engineering best practices.
- Developed and modernized Kotlin microservices to manage files and data.
- Analyse and extract services from a legacy application to create microservices in Kotlin.
- Collaborating in a squad of 20 peoples.
- Deploying and reviewing pull requests on GitHub with Sonar analysis.
- Monitoring applications with Dynatrace and Splunk.

Stack: Kotlin | Java | IntelliJ | Azure | Gradle | GitHub | Sonar | Dynatrace | Splunk

Morgan Stanley

March 2023 - March 2025

Data Scientist, Strategist & Data Analyst

Developed and maintained tools and applications for data manipulation, visualization, and reporting.

- Managed dashboard creation and maintenance for trading desks, providing key insights and performance metrics to facilitate informed decision-making.
- Collaborated with cross-functional teams to gather requirements, design data-driven solutions, and implement scalable data pipelines.

- Utilized KDB/Q for real-time data processing, querying & analysis to support high frequency trading strategies.
- Conducted comprehensive data audits and quality assessments to ensure data integrity and Reliability.
- Generated actionable insights through exploratory data analysis. Developed data-driven recommendations to optimize business processes and mitigate risks.

Stack: Q/KDB+ | Python | BitBucket | TeamCity

FX Innovation

August 2021 - March 2023

Senior Java Developer & Tech Lead

Client: Desjardins. Tech lead on the modernization team. Java/Spring API for Oracle security. Deployment on Weblogic and Microsoft Azure. CI/CD on BitBucket / GitHub and Concourse. Testing with JUnit, Cucumber, Gherkin and Sonar. Project management with Jira and Confluence for documentation.

- Participated in modernizing the architecture.
- Updated and maintained an app to process files with Java/Spring.

Stack: Java | Spring | Swagger | Git | BitBucket | GitHub | Sonar | Junit | Cucumber | Gherkin | Concourse | Postman | Jira

Systematix

September 2019 - August 2021

Senior Fullstack Developer

Client: National Bank of Canada. Maintained apps with NodeJS. Deployment with Ansible and Terraform. Tests with Axios, Mocha and Sonar. App monitoring with Splunk and DataDog. Installation and OpenText Exstream for document generation. CI/CD on BitBucket, Jenkins, Terraform on OpenShift.

- Updated Dockerfiles. Updated deployment scripts with VI on Linux servers.

Stack: NodeJS | FeathersJS | Express | Axios | Mocha | OpenShift | Jenkins | BitBucket | Ansible | Nexus | Vault | Jira | Confluence | Bash | Linux | VI | Docker | Dockerfile | Sonar | Splunk | DataDog | PostgreSQL | Postman | OpenText Exstream | Control-M | MFT

Full stack Developer

Client: McGill University. Maintained a RabbitMQ app with Java and Junit. Added a security layer on API Connect.

- Created an algorithm and an app to confront JSON files with JQuery. Deployed on OpenShift and GitLab.

Stack: Java | RabbitMQ | Junit | JQuery | API Connect | NodeJS | Jira | Confluence

Ippon Technologies, France

November 2018 - August 2019

Full stack Developer | 10 months

Client: BNP Paribas. Maintained a customer account consultation app on agile/scrum process. Testing with JUnit and Cucumber. Testing load on servers with Gatling. Helped two Java junior developer.

- Developed a Java, Spring app. Created producers and consumers Kafka. Upgraded Cassandra data models.

Stack: Java | Spring | JUnit | Cucumber | Gatling | Kafka | Cassandra | Solr | Git | Jira | Bash | Linux servers

CGI, France

September 2018 - November 2018

Client: SFR (Phone operator)

Full stack Developer

Migrate ERP (Enterprise Resource Planning). Migrate Oracle servers. Maintained an app of professional accounts management. Upgraded a Java app (upgrade Java version from Java 6 to Java 8).

Stack: Java | Oracle | Git | Bash

Altitude Infrastructure, France

July 2017 - August 2018

Developer

Application manager for phone operators. Maintained a C# app and developed a new C# app with .NET. Created a Front-end app to visualize KPI with C# and Bootstrap. Installed a new ERP (Enterprise Resource Planning) with transcoding tables. Migrated ERP from Adonix to SageX3. User testing.

Stack: C# | ASP.NET | Bootstrap 3 | JQuery | HTML | CSS | Bash | Git | Redmine | SQLServer

Saagie, France

October 2016 - June 2017

Developer | Intern

Logging app to track customers journey on the website with Java, Hadoop, Cloudera and Hive. Logging javascript library with JQuery. UX/UI improvement.

Stack: Java | Kafka | Hive | Hadoop | Cloudera | Bash | Javascript | JQuery

Personal Projects

Personnal Website and portfolio

2025

A personnal blog / website using NodeJS, Sveltekit, TailwindCSS, Netlify and Flowbite
<https://cleguen.com>

Project "Life Gamera"

2023

The Conway's Game of Life implementation using Raylib and Python.

<https://github.com/CorentinLeGuen/lifegameray>

Project "Lotto 6.49 API"

2022

An API containing Lotto draws since 1982-06-12. Python3, Flask, SQLite, Docker

More here: <https://github.com/CorentinLeGuen/lotto-6-49-api>

Project "Python on Page"

2022

Running Python on browser using WebAssembly. Python3, Flask, WebAssembly

<https://github.com/CorentinLeGuen/python-on-page>

Project "pycasso"

2022

A project to transform a picture into ASCII code using Python and Numpy.

<https://github.com/CorentinLeGuen/pycasso>

Renew project “Le Compte est Bon”

2022

Renew the project from 2013 with new technologies to solve the algorithm using Python

More here: <https://github.com/CorentinLeGuen/le-compte-est-bon>

Project “Scrap and analyse”

2021 & 2022

A project to scrap data from websites and analyse them to predict next incoming update. Project hosted on a VPS (Virtual Proxy Server) on Linux. Python3, Pandas, BeautifulSoup4, Flask, Bash

Projetct “Loto Quebec Last Result API”

2021

A project to scrap Data from Loto 6.49 website using Java

<https://github.com/CorentinLeGuen/LotoQuebec649LastResults>

Project “array list” in Go

2020

A project made in Go to provide a Java arraylist implementation like. This project have been cloned more than 250 times and has 1 fork. Go. Code source here:

<https://github.com/CorentinLeGuen/le-compte-est-bon>

Project “Simple KeyLogger”

2020

A small project to save input keys into a log file. Python3

<https://github.com/CorentinLeGuen/SimpleKeyLogger>

Project “Where is ISS?”

2018

A front-end project to track the ISS position on earth globe. Javascript, JQuery, PlanetaryJS

Technology used: <https://miniature.earth/demo/space-view.htm>

Project “Le Compte est Bon”

2013

Created an algorithm to solve the game and a GUI with GTK. Made with Python2 and PyGTK.

More here: https://fr.wikipedia.org/wiki/Des_chiffres_et_des_lettres#Le_Compte_est_Bon

Education and Professional Development

AWS Certified Developer Associate, Amazon

2025

Learning AI Through Visualization, Columbia University

2025

Data Science & Machine Learning with Python, Udemy Online course at udemy.com

2022

Certificate of Language Training in English, ACA Online course at lesateliers.ca

2021

Specialization: Programming with Google Go, Coursera Online course at coursera.org

2019

Intermediate R Course, DataCamp Online course at datacamp.com

2019

Deep Learning, National Conservatory of “Arts et Métiers”, France Online course

2018

Master degree of Computer Software Engineering, University of Rouen Normandie, France

2018

Licence Diploma of Development, University of Rouen Normandie, France

2016

Introduction to C++ Development, Polytechnic School of Lausanne, Swiss Online course at coursera.org

2013

Introduction to Java Development, Polytechnic School of Lausanne, Swiss Online course at
coursera.org

2013

Languages: Bilingual in English and French